NASRULLAH KAISHER SIJAN



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GitHub: https://github.com/Nasrullah4222

Portfolio: https://nas-portfolio-tau.vercel.app/

PROFILE

Computer Science and Engineering student with hands-on experience developing full-stack and backend applications using JavaScript, React.js, Node.js, Express.js, REST APIs, MongoDB and MySQL. Experienced in authentication and authorization, middleware, routing, database operations, API integration, and Generative AI integration. Seeking Software Engineering Internship opportunities focused on backend, full-stack development, Software Testing, and Quality Assurance. Also seeking job or internship opportunities in Computer Vision and Image Processing.

EDUCATION

East West University, Bangladesh **Graduation: December 2026**
B.Sc. in Computer Science and Engineering CGPA: 3.10/4.00

Relevant Coursework: Software Engineering, Object-Oriented Programming, Data Structures, Algorithms, Database Management System, Structured Programming, Operating Systems, Computer Networks, Cyber Security and Ethics, Information System Analysis and Design

Ideal College, Dhanmondi **2021**
Higher Secondary Certificate (HSC), Science GPA: 5.00/5.00

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++, or comfortable with any other language.

Frontend Development: React.js, HTML5, CSS, Tailwind CSS

Backend Development: Node.js, Express.js, django, API Development, Routing, Middleware, RESTful API, API Security, SQL injection, Docker

Authentication and Security: Authentication, Authorization, JWT, Cookies, Headers

Databases: MongoDB, MySQL, DBMS,

Software Engineering: Object-Oriented Programming, OOP, UML, SOLID Principles, MVC Architecture

AI and Machine Learning: PyTorch, TensorFlow, OpenCV, Transformers

Tools and Deployment: Git, GitHub, Vercel, Render, Kubernetes

PLACES WHERE I CAN CONTRIBUTE

DATABASE MANAGEMENT:

I can contribute to database development, particularly in database architecture, ER diagram design, and SQL query optimization.

API DESIGNING:

With my foundational knowledge of API development and Data Structures Algorithms (DSA), I can

contribute to designing logical API solutions, developing RESTful APIs, testing endpoints using Postman, and implementing essential API security practices.

FRONTEND ASSISTENT:

I can provide additional support to the Frontend Team using my foundational knowledge of HTML, CSS, and React, particularly in UI adjustments, component integration, and basic frontend implementation.

DEVOPS:

I can assist the DevOps Team with basic server configuration, deployment, and infrastructure management using Nginx, Kubernetes, and cloud platforms.

SOFTWARE ENGINEERING:

I can be involved in both Low-Level Design (LLD) and High-Level Design (HLD) activities using my foundational knowledge of software architecture and system design.

AI ML RESEARCH & INTEGRATION:

With strong fundamentals in Image Processing and Computer Vision, I can contribute to any kind of research activities in AI-ML research, model Fine-tuning, and AI Engineering.

SOFTWARE ENGINEERING PROJECTS

Interview Planner *Production-style Individual Project*

- **Engineered** a full-stack interview planning application using React.js for the frontend and Node.js for backend services.
- **Implemented** authentication, authorization, routing, middleware, and frontend-backend data flow.
- **Integrated** Generative AI to generate structured interview preparation plans and analysis.
- **Deployed** the application on Vercel.

Live Demo: https://ai-interview-road-map-analysis-fron.vercel.app/

Flower Classification *Production-style Individual Project*

- **Built** a full-stack web app that uses a Vision Transformer model to classify flower species.
- **Developed** backend APIs with Node.js, Express, FastAPI, and MySQL for database storage.
- **Added** AI-powered web search to give users necessery information about classified flowers.
- **Handled** user authentication and core backend operations for seamless app functionality.

GitHub: https://github.com/Nasrullah4222/FlowerClassification

Banking System *Production-style Individual Project*

- **Developed** a backend-focused banking system using Node.js, Express.js, and MongoDB.
- **Designed and implemented** REST APIs with structured routing, controllers, and backend business logic.
- **Integrated** Nodemailer for transactional email functionality.
- **Implemented** authentication, database operations, and complex controller workflows.

GitHub: https://github.com/Nasrullah4222/Banking-System

BanglaOCR Text Generation *Capstone Project – In Development*

- **Developing** an ML-powered application for Bangla and English handwritten text recognition.
- **Training** machine learning models for handwritten text recognition.
- **Integrating** trained models into a web application to establish an end-to-end OCR workflow.

Live Demo: https://automated-bangla-english-handwritten-6puz.onrender.com

Ustaad – Digital Parking Service *Team and Course Project*

- **Contributed** to a MERN-stack digital parking service.
- **Integrated** Map API functionality for location-based parking features.
- **Collaborated** with team members to integrate and test the mapping functionality.

Live Demo: https://ustaad-application-mern-stack.vercel.app/

ACHIEVEMENTS

- **Semifinalist**, Climate Science Olympiad 2024.
- **Qualified for participation** in the Senior Category of the International Academic Research Competition 2025 by achieving an 80%+ score.
- **Participated** in the Project Showcase segment of National ROBO FEST 2026.

AI & ML PROJECT

Endoscopy Contamination Image Segmentation: Fully Supervised vs. Self-Supervised Models

- **Compared** fully supervised and self-supervised learning approaches for segmenting contamination in endoscopic images.
- **Evaluated** image segmentation models to investigate the effectiveness of self-supervised representations for medical imaging.

GitHub: Endoscopy Contamination Segmentation Models

Flower Classification with Supervised and Self-Supervised Models

- **Compared** supervised and self-supervised learning approaches for flower image classification.
- **Explored** learned image representations for classifying flower species.

GitHub: Flower Classification Models

ACADEMIC PAPER PROJECT

Endoscopy Image Segmentation of Contamination with Self-Supervised Learning

- **Investigated** self-supervised learning approaches for reducing manual image segmentation requirements.
- **Analyzed** endoscopic images for automated contamination segmentation.

Green Technology Adoption Among University Students in Dhaka

- **Investigated** behavioral factors influencing university students' adoption of green technologies.

- **Analyzed** motivations and barriers affecting energy-efficient technology usage.

EXTRACURRICULAR ACTIVITIES

Volunteer, National STEM Olympiad *2023*

- **Supported** STEM-focused activities and **contributed** to engaging students in science, technology, engineering, and mathematics initiatives.

Volunteer, Green Life Society *2020–2022*

- **Promoted** affordable and accessible plants while **supporting** community members facing financial difficulties.

Volunteer, Let's Plant Trees BD *2019–2020*

- **Participated** in tree plantation initiatives to **promote** environmental sustainability and contribute to greener local communities.

LANGUAGES

Bengali: Native **English:** IELTS 6.0 – No individual band below 6.0